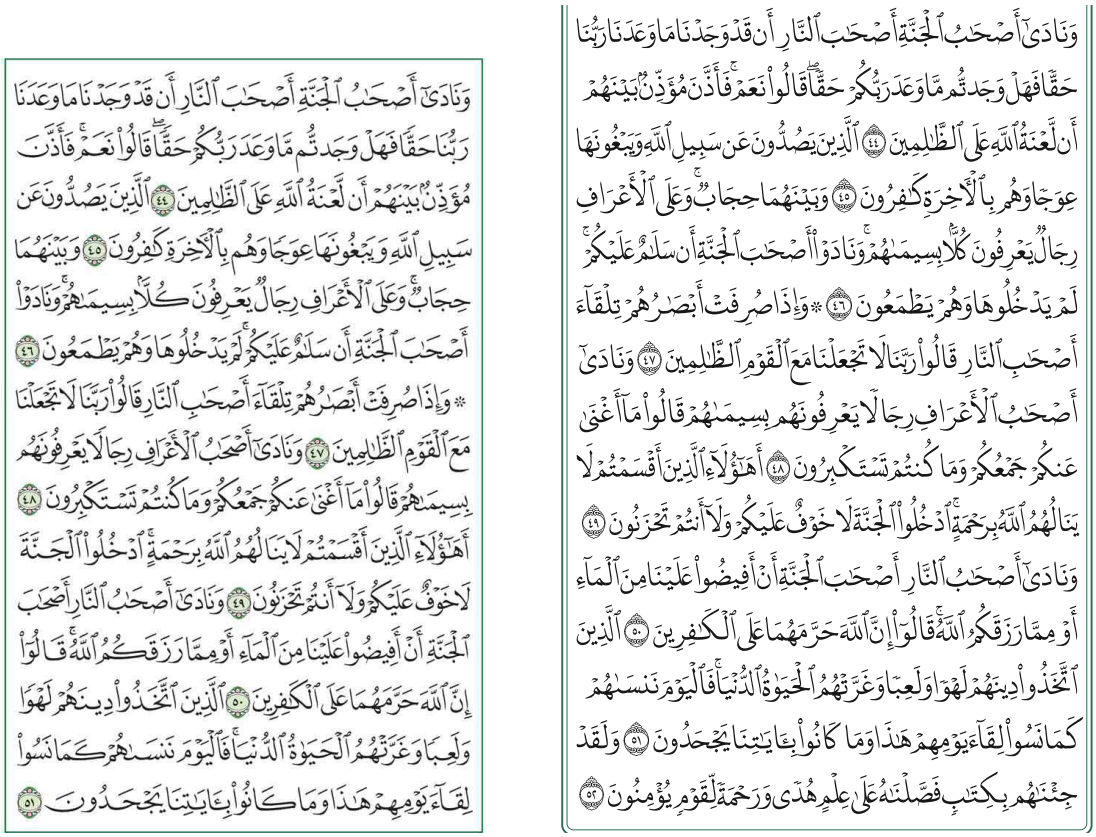
Quran-Nazm: Reducing Layout Variance in Quran Typesetting

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(a) Madina 1441 H print, page 156. **(b) Quran-Nazm, page 156.**

Figure 1: The same passage (Al-A'raf 44 onward) in the Madina print and in Quran-Nazm. In the print, line 11 carries 25% more text than fits at natural size while the lowest lines are spaced out; in Nazm the lines of a page carry equal shares of text.

ABSTRACT

The Madina Mushaf, written by the calligrapher Uthman Taha and printed by the King Fahd Complex, is the reference for most printed Qurans. Its hand is admired; its page layout, assembled manually, is not uniform. Measuring all 8,790 full lines of the current (1441 H) edition against a uniform DigitalKhatt reproduction of the same hand, we find that a line holds between 4 and 15 words, that its natural length ranges from 62% to 136% of the fixed line width, and that one line in three is more than 10% too long or too short. On the printed page, pen width varies by 13% between lines and tracks crowding, and 6.5% of lines open with the marker that closes the previous ayah. We present *Quran-Nazm*, an automatic typesetter that replaces manual layout with explicit rules: every hizb fills exactly ten 15-line pages, every page ends on an ayah, and each hizb is set as one natural line that is cut into equal pages and equal lines by dynamic programming, then justified with DigitalKhatt's letter elongation under a 25% compression limit. The complete Mushaf renders in about one minute to 604 vector pages at 93% of the Madina type size. Within a page, the spread of line stretch falls from 0.063 to 0.039, a reduction of 38%, with every page ending on an ayah and no colliding words.

CCS CONCEPTS

• **Applied computing** → **Document preparation**; *Arts and humanities.*

KEYWORDS

Arabic typography, Quran, Madina Mushaf, line breaking, justification, kashida, DigitalKhatt

1 INTRODUCTION

The Madina Mushaf has shaped how the Quran looks to readers for four decades. Its fifteen-line pages, each beginning and ending on an ayah in most editions, have become a shared map: readers memorise by page and quote by position. The text was written by hand by Uthman Taha and the pages were then composed manually [11]. The hand is the model that digital efforts try to reproduce; the composition is something else. Lines were filled by what fitted, stretched or compressed by eye, and in places the written words were scaled to close a line.

Digital Arabic typography has meanwhile reached the point where this hand can be set by rule. DigitalKhatt [1, 2] models the Madina hand as an OpenType font whose letters can elongate (kashida and alternate forms), and HarfBuzz [9] shapes it on any platform. Earlier work in the TEX community established justification by elongation for Arabic [3, 8] and typesetting of traditional Quranic text [6]. Our own renderer, quran-madina-html [4], reproduces the existing Madina layouts line for line.

This paper asks a different question: *if a machine sets the Mushaf, which rules should it follow so that every page is as harmonious as the hand deserves?* Our contributions are:

(1) a quantitative account of layout variance in the Madina 1441 H print: line density, pen weight and ayah-marker placement (Section 2);
(2) an analysis of the design space, from the unit of consistency to the compression limit (Section 3);
(3) Quran-Nazm, an open implementation that renders the full Mushaf to PDF under explicit rules (Section 4), and its evaluation against the print (Section 5).

2 THE PROBLEM: VARIANCE IN MANUAL SETTING

We use the 1441 H edition throughout, the current one. Two sources are measured: the line breaks of the print, from the DigitalKhatt layout data used by quran-madina-html [4], and the printed pages themselves, from the King Fahd Complex PDF [12].

2.1 Line density

Each full line of the print was shaped once, with no justification, in the QMH New Madina font (a patched DigitalKhatt font), and its natural length compared with the fixed line width. If the hand were set evenly, all lines would sit near 100%. Fig. 3 shows they do not: natural length runs from 62% to 136%; 15% of lines are more than 10% short and had to be stretched, and 18% are more than 10% long and had to be compressed. A full line holds 4 to 15 words and 21 to 52 letters.

Word count alone does not explain this. On page 72, lines 10, 11 and 12 each hold nine words, yet their natural lengths are +11%, −22% and +14%: word length matters as much as word count. Page 156 (Figs. 1(a) and 2) is the most uneven full page: most of its lines are overfull, up to +25% on line 11, while its last line is 15% short, so the same hand appears compressed at the top of the page and spaced out at the bottom. Table 1 summarises the measurements of this section.



Figure 2: Page 156 of the print, each line set once at its natural length in the DigitalKhatt reproduction of the hand. Dashed rules mark the printed line width; red shading is text that had to be compressed into it, green the gap that had to be filled by stretching.

Table 1: Layout variance measured on the Madina 1441 H print.

Measure	Value
Full-width lines measured	8,790
Words / letters on a full line	4–15 / 21–52
Natural length vs. line width	62–136%
Lines off by more than 10%	33%
Pen width, light vs heavy line (5th–95th pct.)	+13%
Pen width spread inside one page (median)	9%
Ink per letter, sparse vs crowded lines	1.36×
Lines opening with an ayah marker	573 (6.5%)

2.2 Pen weight and texture

On the scanned pages (720 × 1123 px, about 100 ppi) we split each page into its 15 line bands, masked the coloured ayah markers, upsampled each band four times and measured the mean pen width

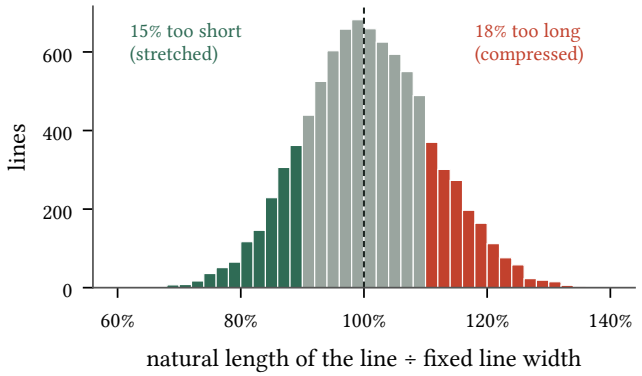


Figure 3: Natural length of the 8,790 full lines of the 1441 H print, relative to the line width. Coloured bars are lines more than 10% off.

as ink area divided by skeleton length. Over 7,320 lines on the 488 pages made only of text:

- a heavy line has a 13% thicker stroke than a light one (5th to 95th percentile), and the spread inside one page is typically 9%;
- on page 102, line 5 is 16% heavier than line 4 directly above it, and page 561 is 19% heavier than page 563 (Fig. 4(a,b));
- weight follows crowding (Fig. 5): sparse lines, which were stretched, are written with 6% thicker strokes than crowded ones, and ink per letter spans 1.36× and correlates with crowding at $r = -0.61$. On page 192 a five-word line carries 65% more ink per letter than a ten-word line of the same page (Fig. 4(c)).

The likely mechanism is scaling of written words to fit: a line scaled up gains stroke weight as well as length. At this resolution a stroke is about two pixels wide, so we read these figures as relative differences; the crops show that they are visible.

2.3 Ayah markers cut off from their ayah

In 573 lines (6.5%) on 362 of the 604 pages, a line opens with the marker that closes the previous ayah, so the ayah ends at the far left of one line and its marker appears at the far right of the next (Fig. 4(d)). All 440 such lines on the text-only pages were confirmed on the scan. This is an accepted convention of the print; we list it because a typesetter with more freedom can avoid it.

3 ANALYSIS AND APPROACH

3.1 Choosing the unit of consistency

Readers value predictable positions: where a section starts, how many pages it spans. Our first target was every rub' (a quarter hizb) on exactly two pages. Table 2 shows why that fails: rub' lengths differ by 2.19×, so the longest rub' forces type down to 56% of the Madina size while the shortest must stretch 2.26× on average. Variance shrinks as the unit grows; the hizb (half a juz) is the smallest unit at which stretching stays within what DigitalKhatt renders gracefully. Hizb lengths still differ by 1.35× (Fig. 6). Ten pages per hizb keeps Madina's own density of about nine words per line; eight pages gives the same stretch but type at 63%. Juz 30, with 37 short surahs and their headers, is a declared exception.

Table 2: Fixed pages per unit, one common line width, no compression. Stretch is the average a unit's lines need.

Unit → pages	Line width vs Madina	Median stretch	Worst stretch
Rub' → 2	1.78×	1.42×	2.26×
Half hizb → 4	1.61×	1.30×	1.68×
Hizb → 8	1.51×	1.22×	1.42×
Juz → 20	1.14×	1.15×	1.22×

3.2 From one line to pages and lines

Two constraints come from the tradition: a page ends at the end of an ayah, and no line starts with an ayah marker. Within them, the layout of a hizb follows one idea: *set the whole hizb as a single natural line, then cut it.* Let the hizb consist of ayahs $1..n$ with natural widths a_k (shaped words plus spaces, each marker glued to its preceding word), and let a surah header count as $h_k \in \{0, 1, 2\}$ lines of width q. With cumulative widths $V_k = \sum_{i \le k}(a_i + h_i q)$ and target $\bar{V} = V_n/10$, the page cuts $0 = c_0 < c_1 < \cdots < c_{10} = n$ at ayah ends minimise

$$\sum_{p=1}^{10} \left(V_{c_p} - V_{c_{p-1}} - \bar{V}\right)^2 + \sum_{p} \pi(c_p), \tag{1}$$

where π penalises a cut inside a surah short enough for one page. Each page's words are then cut into its $L = 15 - H$ text lines (after H header lines) by minimising $\sum_\ell (w_\ell - \bar{w})^2$ over word boundaries, with every surah starting on a new line. Both problems are solved exactly by dynamic programming, in the spirit of Knuth and Plass [13], with a sum-of-squares cost.

Choosing cuts *together* matters. Cutting each page and each line at its nearest boundary independently left pages nearly empty beside long ayahs (worst line 3.40×) and lines up to half a word off target; jointly chosen cuts bring the worst line to 2.16× and the spread inside a page from 0.057 to 0.039.

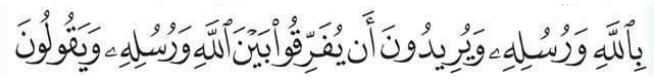
3.3 Line width and compression

The whole book shares one line width $W = (1 - \sigma)\max_\ell w_\ell$, where w_ℓ are the natural line widths of hizbs 1–58 and σ is the compression limit. A line is then stretched by $s_\ell = W/w_\ell$, or compressed when $s_\ell < 1$. The limit sets type size and where stretch is centred, not evenness (Table 3). Squeezed lines are those where tighter spaces were not enough and letters had to be scaled horizontally, a distortion the print never shows. We chose $\sigma = 25\%$: type at 90% of Madina in this model, with squeezing still limited.

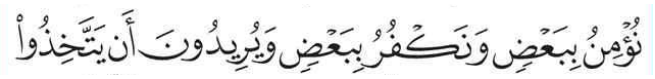
4 IMPLEMENTATION

Quran-Nazm [5] is a single Python program, `render_mushaf.py`, built on the data and justifier of quran-madina-html [4]. Fig. 8 shows its stages.

Text and shaping. The Hafs text of the 1441 H layout data [4, 15] is split into ayahs and words; rub' and hizb boundaries come from

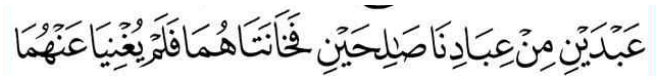


p. 102, line 4: pen 2.11 px, 93 px^2 ink per letter



p. 102, line 5: pen 2.45 px (+16%), 122 px^2 ink per letter

(a) Neighbouring lines written with different pens.



p. 561, line 9: pen 2.85 px, the heaviest page measured

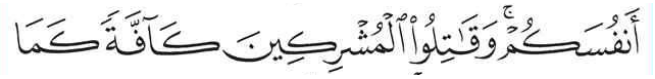


p. 563, line 9: pen 2.14 px, two pages later

(b) The same line slot on nearby pages.



p. 192, line 1: 10 words, natural length 97%, 89 px^2 ink per letter



p. 192, line 14: 5 words, natural length 67%, 147 px^2 ink per letter (+65%)

(c) One page, two textures: a sparse line is stretched, enlarged and darkened.



p. 567, lines 2–3: the last word of ayah 11 (left box) and its marker (right box)

(d) An ayah's marker carried to the next line.

Figure 4: Anomalies in the Madina 1441 H print: crops of the King Fahd Complex PDF [12] at native resolution (about 100 ppi). Measurements are those of Section 2.

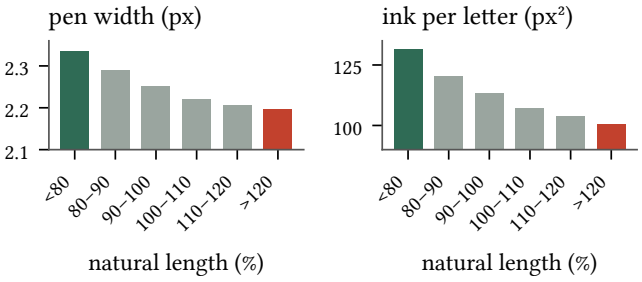


Figure 5: Pen width and ink per letter on the scan, grouped by how crowded the line is (its natural length).

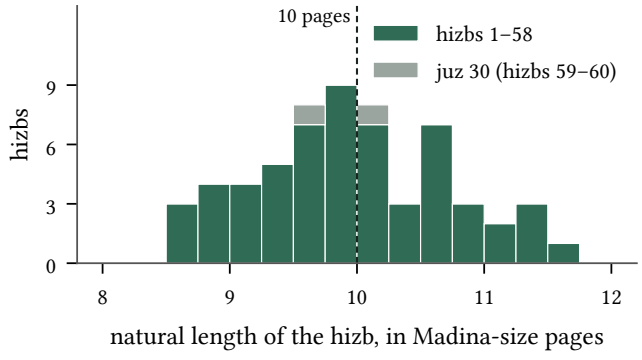


Figure 6: Natural length of the 60 hizbs at Madina type size. Fixing every hizb to ten pages means absorbing this spread.

quran-meta [14]. Every word is shaped once with HarfBuzz in the QMH New Madina font.

Table 3: Compression limit σ (hizbs 1–58, analysis model).

σ	Type vs Madina	Median stretch	Worst stretch	Lines squeezed	Worst squeeze
10%	75%	1.34×	1.94×	0%	4%
15%	79%	1.26×	1.83×	1%	9%
20%	84%	1.19×	1.73×	3%	14%
25%	90%	1.12×	1.62×	8%	20%
35%	103%	0.97×	1.40×	41%	30%
45%	122%	0.82×	1.19×	89%	41%

Juz 30. Its two hizbs share exactly 24 pages, bringing the book to 604 pages, the count of the print. Short surahs each start on a new line under a two-line header, so text width is a poor guide to fullness; a candidate page is scored by the lines it really needs at width W. No page may need more than the compression limit or be sparser than 1.35× average stretch; a split surah keeps at least two ayahs on each page.

Justification. Lines that are too short are stretched with the DigitalKhatt elongation rules [1] as ported in quran-madina-html. Lines that are too long tighten their spaces to at most 60% and are then scaled horizontally. The font's word-level shrink features were dropped: shaped inside a full line, they pushed words into their neighbours.

Ink check. Every set line is tested for ink contact between neighbouring words: glyph boxes that intersect are rasterised and compared pixel by pixel. Ten contacts in the book, nearly all a small pause mark meeting the tall stroke of a following kaf, are cleared by widening only that space.

Output. A minimal PDF writer stores each glyph outline once as a form object shared by all pages and places it by matrix. Surah headers are vector art (a ribbon, a cartouche with a double outline,



(a) Page 591: Al-Tariq and Al-A'la.

(b) Page 2: Al-Baqara.

(c) Page 1: Al-Fatiha, Al-Baqara.

Figure 7: Quran-Nazm pages. Each surah begins on a new line under its header; lines of a page carry equal text; every page ends on an ayah.

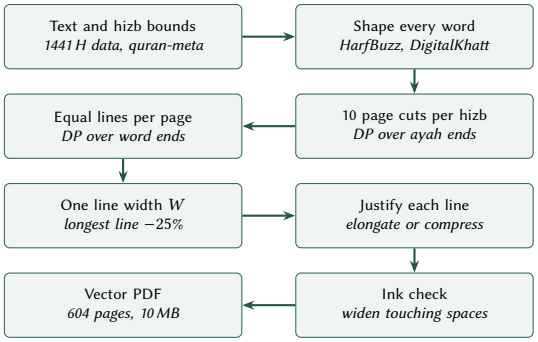


Figure 8: The Quran-Nazm pipeline.

Table 4: Madina 1441 H print and Quran-Nazm.

	Madina	Nazm
Pages	604	604
Pages per hizb	varies	10 (juz 30: 24)
Type size	100%	93%
Spread of stretch inside a page	0.063	**0.039**
Lines more than 10% compressed	16%	5%
Median line stretch	1.00×	1.07×
Lines opening with a marker	6.5%	0
Pen weight across lines	±13%	uniform
Lines squeezed horizontally	none	21% (max 24%)

eight-pointed stars); running heads use Amiri Quran [10]. The whole Mushaf renders in about a minute.

5 RESULTS AND COMPARISON

Table 4 compares the render with the print. The key measure of harmony is the spread of stretch *inside a page*: the standard deviation of $\log s_\ell$ over a page's lines, taken as the median over pages. It falls from 0.063 in the print to 0.039 in Nazm (Fig. 9), and the average stretch of neighbouring pages differs slightly less than in the print (median 0.044 vs 0.051 in log stretch). Every page ends on an ayah, no line opens with a marker, and no two words touch.

The price is visible in two places. Nazm stretches more on average (median 1.07×): fixing every hizb to exactly ten pages sets the line width by the densest text in the book, so the lighter hizbs must stretch. We accept this shift, since the aim of the work is the variance within a page, not the mean; and at the 25% limit, one line in five is compressed partly by horizontal scaling, most visibly on the densest hizb (Al-Mujadila). Both trace to the compression side of the font: DigitalKhatt elongates gracefully but has few safe ways to contract a word inside a line. In juz 30, keeping every page evenly filled splits five short surahs across pages (each with at least two ayahs per page), in exchange for pages between 0.96× and 1.25× stretch (Fig. 7).

6 DISCUSSION AND FUTURE WORK

Three limits remain. First, compression: a set of compression forms in DigitalKhatt that is safe inside a full line (narrower alternates, tighter ligatures, shorter tails) would remove horizontal scaling and

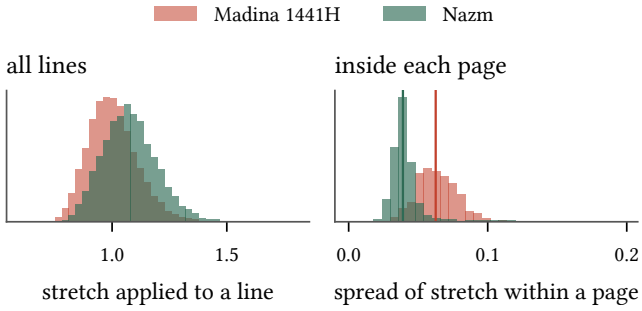


Figure 9: Stretch applied to each line (left) and its spread within each page (right), print vs Nazm. Vertical lines mark the medians. The main aim, a lower variance of stretch within each page, is achieved (right: 0.063 → 0.039). The slight shift towards more stretching overall (left) is the cost of the extra constraint of exactly ten pages per hizb, and is accepted.

allow Madina-sized type. Second, the gap between hizbs: fixing every hizb to ten pages leaves a gradual difference in density between distant hizbs (standard deviation 0.085 of log stretch), which never shows within a spread. Third, measurement: pen-weight figures would sharpen with a 300 ppi scan. Surah headers and the treatment of juz 30 are design choices open to calligraphers' review.

7 CONCLUSION

The Madina Mushaf's layout varies by much more than its hand: a third of its lines are off their natural length by over 10%, and stroke weight follows the crowding. Quran-Nazm shows that the same hand, set by explicit rules, can hold every hizb to ten pages, end every page on an ayah, and cut the variation of lines within a page by 38%. The program, data, measurements and the rendered Mushaf are available under a dual Waqf 2.0/commercial licence [5].

ACKNOWLEDGMENTS

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